BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Connectedness

Separated Sets, Connected Sets, Components

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Conducted by

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Separated Sets

Separated Sets

Let X be a metric space. Two subsets $A, B \subseteq X$ are said to be *separated* if

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Connected and Disconnected Sets

Connected Sets

A subset $A \subseteq X$ of a metric space X is said to be *connected* if it cannot be expressed as the union of two non-empty, separated subsets.

Disconnected Sets

A subset $A \subseteq X$ of a metric space X is said to be *disconnected* if it can be expressed as the union of two non-empty, separated subsets.

Disconnected Space and Clopen Sets

A metric space X is disconnected if and only if there exists a non-empty, proper subset $U \subset X$ such that U is both open and closed in the topology on X.

Disconnected Sets and Clopen Sets

A subset $A \subseteq X$ of a metric space X is disconnected if and only if there exists a non-empty, proper subset $U \subset A$ such that U is both open and closed in the subspace topology on A.

Proof:

($\Rightarrow$) Suppose A is disconnected. Then there exist non-empty separated sets B and C such that $A = B \cup C$. We will show that B is a non-empty proper subset of A that is both open and closed in the subspace topology on A. Since B and C are separated, we have $\overline{B} \cap C = \emptyset$ and $B \cap \overline{C} = \emptyset$. This implies that B is closed in A because $\overline{B} \cap C = \emptyset$ means that $\overline{B} \subseteq A \setminus C$, and B is open in A because $B \cap \overline{C} = \emptyset$ means that $B \subseteq A \setminus \overline{C}$. Thus B is a non-empty proper subset of A that is both open and closed in the subspace topology on A.

($\Leftarrow$) Conversely, suppose there exists a non-empty proper subset $U \subset A$ that is both open and closed in the subspace topology on A. We will show that A is disconnected. Since U is open in A, there exists an open set V in X such that $U = V \cap A$. Since U is closed in A, we have $\overline{U} \cap A = U$. Let $W = A \setminus U$. Then W is non-empty because U is a proper subset of A. We will show that U and W are separated. First, we have

$$\overline{U} \cap W = (\overline{U} \cap A) \cap W = U \cap W = \emptyset.$$

Next, we have

$$U \cap \overline{W} = U \cap (A \setminus U) = \emptyset.$$

Thus U and W are separated, and $A = U \cup W$ is a union of two non-empty separated subsets. Therefore A is disconnected.

Example of a non-trivial Disconnected Set

Consider the subset of $\mathbb{R}^2$ defined by

$$S = \left\{ \left(x, \sin\left(\frac{1}{x}\right) \right) : x \neq 0 \right\}.$$

Show that S is disconnected.

Connectedness and Continuous Functions

Continuous Image of Connected Set

If $f : X \to Y$ is a continuous function between metric spaces and $A \subseteq X$ is connected, then $f(A)$ is connected in Y.

Union of Connected Sets

Union of two Connected Sets

The union of two connected sets, having non-empty intersection, is connected.

Proof: Let A and B be two connected sets such that $A \cap B \neq \emptyset$. We will show that $A \cup B$ is connected. Suppose, for contradiction, that $A \cup B$ is disconnected. Then

$$A \cup B = U \cup V,$$

where U and V are non-empty separated subsets.

Since A is connected, it cannot meet both U and V. Indeed, if both $A \cap U$ and $A \cap V$ were non-empty, then

$$A = (A \cap U) \cup (A \cap V)$$

would express A as a union of two non-empty separated subsets, because separatedness is inherited by subsets. Hence $A \subset U$ or $A \subset V$. Similarly, $B \subset U$ or $B \subset V$.

Choose a point $p \in A \cap B$. The point p lies in exactly one of U and V, since separated sets are disjoint. If $p \in U$, then the connectedness argument forces both $A \subset U$ and $B \subset U$. Thus $A \cup B \subset U$, so V is empty, a contradiction. The case $p \in V$ is identical. Therefore $A \cup B$ is connected.

Arbitrary Union of Connected Sets

Arbitrary union of connected sets, having non-empty intersection, is connected.

Proof: Let $\{A_\alpha\}_{\alpha\in I}$ be a family of connected sets such that $\cap_{\alpha\in I} A_\alpha \neq \emptyset$. We will show that $\cup_{\alpha\in I} A_\alpha$ is connected.

Suppose, for contradiction, that $A = U \cup V$ is a separation. Since $p \in A$, either $p \in U$ or $p \in V$. Assume $p \in U$; the other case is symmetric.

For each α, the set A_α is connected and contains $p \in U$. If A_α met V, then

$$A_\alpha = (A_\alpha \cap U) \cup (A_\alpha \cap V)$$

would be a separation of A_α, because both parts would be non-empty and separated. This is impossible. Hence $A_\alpha \subset U$ for every α. Therefore

$$A = \bigcup_{\alpha\in\Lambda} A_\alpha \subset U,$$

so $V = \varnothing$, contradicting the assumption that V is non-empty. Thus A is connected.

Connectedness of Topologist Sine Curve

The *topologist sine curve* is the subset of $\mathbb{R}^2$ defined by

$$T = \left\{\left(x, \sin\left(\frac{1}{x}\right)\right) : x > 0\right\} \cup \{(0, y) : -1 \leq y \leq 1\}.$$

Show that T is connected.

Theorem

A non-empty subset X of $\mathbb{R}$ (with usual metric) is connected if and only if X is an interval or a singleton.

Proof:

A singleton is connected because it cannot be written as the union of two non-empty disjoint sets.

Now let $I \subset \mathbb{R}$ be an interval with at least two points. Suppose, for contradiction, that I is disconnected. By the clopen characterization, there is a non-empty proper subset $U \subset I$ which is both open and closed in the subspace topology on I. Choose

$$u \in U, \qquad v \in I \setminus U.$$

If necessary, interchange the names of u and v, or replace the following argument by its mirror image, so assume $u < v$.

Because I is an interval, $[u, v] \subset I$. Consider

$$E = U \cap [u, v].$$

Then E is non-empty, because $u \in E$, and E is bounded above by v. Let

$$c = \sup E.$$

Then $u \leq c \leq v$.

Since U is closed in I, and points of E can be chosen arbitrarily close to c from below, we get $c \in U$. Since $v \notin U$, it follows that $c < v$.

Because U is open in I, there exists $\varepsilon > 0$ such that

$$(c - \varepsilon, c + \varepsilon) \cap I \subset U.$$

Choose a number t with

$$c < t < \min\{v, c + \varepsilon\}.$$

This is possible because $c < v$. Since $[u, v] \subset I$, we have $t \in I$, and the displayed inclusion gives $t \in U$. Also $t \in [u, v]$, so $t \in E$. But $t > c = \sup E$, a contradiction.

Therefore no such non-empty proper clopen subset U exists, and I is connected.

Theorem

The Real line $\mathbb{R}$ is connected.

Proof: The real line is an interval, so it is connected by the previous theorem.

Generalized Intermediate Value Theorem

Generalized Intermediate Value Theorem

Let X be a connected metric space and $f : X \to \mathbb{R}$ be a continuous function. If $f(x_1) < c < f(x_2)$ for some $x_1, x_2 \in X$ and $c \in \mathbb{R}$, then there exists some $x \in X$ such that $f(x) = c$.

Components of a Metric Space

Component

Let X be a metric space. A *component* of X is a maximal connected subset of X. In other words, a component is a connected subset of X that is not properly contained in any other connected subset of X.

Properties of Components

Properties of Components

Let X be a metric space and let $\{C_\alpha\}_{\alpha \in A}$ be the collection of all components of X. Then:

1. Components are closed sets.
2. Components need not be open sets.
3. Components of a metric space are either identical or pairwise disjoint.
4. The union of all components of X is X itself, i.e., $\cup_{\alpha \in A} C_\alpha = X$.
5. Each component is a connected subset of X.
6. If X is connected, then X itself is the only component of X.

Components of $\mathbb{Q}$

Find the components of the metric space $(\mathbb{Q}, d)$, where d is the usual metric on $\mathbb{R}$.

Thank You!

We'd love your questions and feedback.

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References

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